

Multiply Gleeful numbers

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Abstract. For positive integers k and n let $f_k(n)$ be the number of ways of representing n as a sum of k powers of consecutive primes. A number is called *k-gleeful* if $f_k(n) > 0$ and *multiply gleeful* if $f_k(n) > 1$ or $f_k(n)f_{k'}(n) > 0$ for some positive integers $k < k'$. Under Schinzel's hypothesis H, we show that there are infinitely many positive integers n such that $f_2(n)f_4(n) > 0$. Under the same assumption we show that $\limsup_{n \rightarrow \infty} f_2(n) = \infty$. This gives a conditional proof of a stronger version of a conjecture of Moore and Sorenson from the preprint [1].

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Keywords. Prime numbers, Schinzel's Hypothesis H , polynomials.

References

- [1] S. Moore and J. P. Sorenson, “Explicit bounds and parallel algorithms for counting multiply gleeful numbers”, (2025), arXiv: 2507.09012v1.